AYISHIK DAS

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Education

- **Academy of Technology (Under MAKAUT/WBUT)** Kolkata, India
  Bachelor of Technology - Computer Science Engineering and Business Systems ; CGPA: 8.28 Oct 2021 - June 2025
  Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Databases, Artificial Intelligence, Machine Learning

Skills Summary

- **Languages** C++, C, Python, Java, JavaScript, Shell
- **Backend/Systems** Node.js, Express.js, FastAPI, REST APIs, Linux, WSL, Make
- **Frontend/Tools** React.js, Next.js, Tailwind CSS, Git, GitHub, Docker, Postman
- **Databases** PostgreSQL, MySQL, SQLite, Redis, pgvector
- **AI/ML** Generative AI, RAG, Vector Search, HNSW

Experience

- **System Engineer** Kolkata, India
  Tata Consultancy Services Jan 2026 – Present
  - Engineered **3 Generative AI PoCs** for enterprise clients, including **2 solutions** involving conversational AI, workflow automation, and enterprise data integration.
  - Developed AI-driven solutions using **Generative AI, Claude, and Python libraries**, integrating with **PeopleSoft** to automate enterprise workflows and data processing.
  - Automated targeted enterprise workflows, reducing manual processing time by approximately **70%** and improving operational efficiency.

Projects

- **FluxTrade (C++20, CMake, Lock-Free Queues, Order Book, Google Benchmark)** Built a deterministic electronic exchange simulator in **Modern C++20** featuring a price-time priority limit order book and zero-allocation matching engine. Implemented **preallocated object pools, intrusive data structures, lock-free SPSC/MPSC queues, and cache-line optimization**, achieving **68.9 ns single-trade matching** and **161.5 ns multi-level sweep** latency in LTO release benchmarks.
  GitHub: FluxTrade *July 2026 – Present*
- **ModelLogger (FastAPI, PostgreSQL, pgvector, Redis, Next.js, Generative AI)** Built a research observability platform for AI/ML teams using **FastAPI, PostgreSQL + pgvector, Redis, and Next.js**, enabling experiment tracking and semantic retrieval of research activity. Implemented **real-time WebSocket streams, background workers, anomaly detection, and AI-generated postmortems**, and validated failure detection and recovery workflows on real **GPT-2** training workloads.
  GitHub: xlogger **Live:** xlogger.vercel.app *May 2026 – Present*
- **Recallix (C++, Python, SQLite, HNSW, Generative AI, RAG)** Built a high-performance persistent memory engine for AI agents with **SIMD-accelerated HNSW indexing**, supporting **1M memory nodes** with **0.106ms average C++ search latency**. Implemented asynchronous index construction, SQLite WAL-based persistence, hybrid neural-heuristic reranking, and cross-model memory transfer achieving **100% retrieval fidelity**.
  GitHub: Recallix **Live:** recallix.vercel.app *Jan 2026 – Apr 2026*
- **PayP (Node.js, PostgreSQL, Sequelize, Express.js, JWT, bcrypt)** Designed and built a **UPI-inspired payment API** exposing reusable endpoints for user authentication, balance management, payment transfers, transaction history, and project/product payments. Implemented **JWT authentication, bcrypt credential hashing, payment PIN protection, and 30-minute account lockouts** against repeated authentication failures.
  GitHub: PayP **Live:** payp.onrender.com *Jan 2025 – May 2025*

Awards and Certifications

- Secured **2nd Place** in **TCS AI Friday**, Kolkata
- Winner of **TechQuisitive 5.0 (Quiz Contest)** and **CxthSense 2.0 (Coding Contest)** at College
- Achieved **5 star ratings in C++ and Java** on **HackerRank**
- Open-source Mentor at **GirlScript Summer of Code** and **Winter of Blockchain**
- **Postman API Fundamentals Student Expert Certificate**
- **Linux Command Line: From Zero to Hero Certificate**

Volunteer Experience

- **Member of IEI Student's Chapter CSBS** Aug 2023-May 2025
  Organizing and hosting different events in college
- **Event Coordinator - TechFiesta[Tech Fest of AOT]** Apr 2024
  Hosted **TechFest of AOT** and **event coordinator of Maths** event.

HOBBIES

- Reading Story Books, Photography, Image Editing(Learning),Watching Cricket and Football matches.